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THE IMPACT OF ICT INVESTMENTS ON EMPLOYMENT IN OECD COUNTRIES: A SECTORAL ANALYSIS

Relatore: Chiar.mo Prof. Jacopo Bonan

Laureando:
Gabriele Ferraresi

Matricola n. 736903

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INTRODUCTION

In the contemporary era, Information and Communication Technology (ICT) has become a cornerstone of economic and social development, influencing various sectors, including employment. The rapid advancement and integration of ICT into business operations, educational frameworks, and daily life have revolutionized the labour market. This study aims to explore the intricate relationship between investment in ICT and the employment, analysing how technological progress from 2000 to 2019 impacts on four different sectors: Manufacturing, Information and communication, Financial and insurance activities, and Professional, scientific and technical activities; administrative and support service activities.

There are two reasons why these activities were chosen: the first rationale is based on the "*OECD Digital Economy Outlook 2024 (Volume 1)*"[1] published by OECD on May 2024. This report provides an analysis of the current state of the digital economy, focusing on how the adoption of ICT technologies is transforming various sectors by improving efficiency, and requiring new skill. In particular, the paragraph in question deals with Internet of Things (IoT) which connects physical objects to Internet, allowing them to collect and exchange data and there is reported that the adoption of Internet of Things (IoT) is far advanced in financial and insurance activities. IoT adoption is remarkably evenly distributed across sectors, ranging from an average of 21% in administrative and support service activities to 31% in transport and storage. As expected, adoption is above average in sectors that produce and move physical objects and below-average in white-collar service sectors such as real estate and administrative and support services. This article helps to identify the sectors that are more likely to gain benefit from ICT investments and those less likely to do so.

The second reason is more practical: working with highly specific data filtered by topic, country, year, and economic activities required me to choose sectors that would yield the most comprehensive data. The reference database is OECD Data Explorer, and these four economic activities provided the most complete datasets.

The importance of investing in the latest technologies

The advent of the digital age has significantly transformed how businesses operate and compete. ICT encompasses a wide range of technologies, including the internet, mobile devices, cloud computing, artificial intelligence, and big data analytics. These technologies enhance productivity, efficiency, and innovation but they also pose challenges, such as the displacement of traditional jobs and the need for new skill sets.

Understanding the relationship between ICT and employment is pivotal for navigating the future of work. This study aims to provide comprehensive insights that can inform decision-making processes at various levels, from government policy to corporate strategy and educational curriculum development. By identifying the opportunities and challenges presented by ICT, stakeholders can better prepare for and shape the evolving labour market, ensuring sustainable economic growth and employment resilience.

There may be mixed results regarding the impact of ICT on employment. On one hand ICT have high potential to create new job opportunities, particularly in tech-driven industries. On the other hand, there is a risk of job losses in sectors vulnerable to automation. Understanding these dynamics is crucial for policymakers, businesses, and educational institutions to develop strategies that maximize the benefits of ICT while mitigating its adverse effects.

Investments in Information and Communication Technology (ICT) play a crucial role in shaping employment, both directly and indirectly, and this connection can be understood through several key mechanisms:

- **Productivity and efficiency gains**

Investments in ICT often lead to significant productivity and efficiency gains across various sectors. When businesses integrate advanced ICT tools such as automation software, cloud computing, and data analytics, they can perform tasks more efficiently and with greater accuracy. This increased productivity can lead to business growth, expansion, and the need for more employees to manage increased operations, thus having a positive impact on employment rates.

For example, a company that invests in an advanced customer relationship management (CRM) system can manage a larger customer base more efficiently, potentially leading to the hiring of more customer service representatives, sales staff, and IT support personnel to manage the business expansion.

- **Creation of new jobs categories**

ICT investments often result in the creation of entirely new job categories and industries. The development and deployment of emerging technologies create demand for specialized skills and roles. Fields such as cybersecurity, data science, software development, and digital marketing have emerged as direct results of advancements in ICT. These new job categories not only provide employment opportunities but also drive innovation and economic growth.

For instance, the rise of big data analytics has created demand for data scientists and analysts who can interpret and utilize large datasets to inform business decisions. Similarly, the growth of e-commerce has led to a higher demand for digital marketers and web developers.

- **Enhanced global competitiveness**

Businesses that invest in ICT can improve their global competitiveness by adopting technologies that enhance their operations and product offerings. This competitive edge can lead to market expansion and increased revenues, which in turn can create more jobs. Companies that leverage ICT to innovate and stay ahead of global trends are often better positioned to capture new markets and grow their workforce.

For example, a manufacturing firm that invests in innovative automation technology may be able to produce goods more efficiently and at a lower cost, allowing it to compete more effectively in international markets. This success can lead to the hiring of additional staff to manage increased production and distribution demands.

- **Improved labour market flexibility and remote work**

ICT investments have revolutionized the way work is performed, particularly by facilitating remote work and flexible work arrangements. The proliferation of high-speed internet, collaboration tools, and virtual meeting platforms enables employees to work from anywhere, reducing geographic barriers to employment. This flexibility can lead to higher employment rates as more people are able to participate in the labour force, including those who may have previously been excluded due to location, disability, or family responsibilities.

For instance, a company that invests in robust telecommuting infrastructure can hire talent from across the globe, drawing on a diverse pool of candidates and potentially increasing its overall workforce.

- **Skill development and workforce adaptation**

Investments in ICT often necessitate the development of new skills among the workforces. As companies adopt innovative technologies, they invest in training and development programs to equip their employees with the necessary skills to utilize these technologies effectively. This continuous upskilling can lead to a more adaptable and resilient workforce, capable of taking on new roles and responsibilities as the technological landscape evolves.

For example, a company that introduces AI-driven tools may provide training programs for employees to learn how to operate and manage these tools, thereby enhancing their skill sets and employability.

- **Economic spillover effects**

The positive impact of ICT investment on employment is not limited to the sectors directly involved in technology. The benefits often spill over into other sectors, creating a multiplier effect. For instance, the growth of the tech industry can drive demand in supporting industries such as education, real estate, and retail, leading to job creation in these areas as well.

For example, a thriving tech industry in a city can lead to increased demand for housing, education, and entertainment, thereby creating jobs in construction, teaching, and hospitality.

To sum up all these points, investments in ICT has a multifaceted impact on employment, driven by these points. By fostering an environment where technology and human capital can grow synergistically, ICT investments contribute to a dynamic and robust labour market.

Literature

Understanding the relationship between the ICT investments and the employment is a topic that has garnered significant interest in recent years. Those who are sceptical about the positive effects of the newest technologies on employment argue that jobs requiring lower skills or routine cognitive tasks are particularly vulnerable to automation, potentially leading to widespread job displacement.

Focusing on manufacturing sector, the Deloitte Insights article "*Negative impacts of technology in the workplace*"[2] (2023) points out that the automation of manufacturing processes through advanced technologies can lead to significant job losses, as machines and software increasingly replace manual labour, making certain roles redundant. While technology boosts productivity and efficiency within manufacturing operations, this does not necessarily translate into job creation. In this sector the productivity gains are often accompanied by a reduction in workforce numbers, which emphasizes the need for industry stakeholders to develop strategies that balance technological advancement with workforce stability. This includes investing in retraining programmes for displaced workers and fostering a workplace culture that prioritises employee mental health amidst technological changes.

In the article "*The Dangers of Automation*"[3] by Michael Chui et al. (2017), the authors explore the implications of increasing automation within various sectors. In particular, it highlights substantial risks to employment in the information and communication sector, since automation can lead to significant gains in efficiency and productivity. As ICT advances, roles traditionally filled by humans, such as customer support and data analysis, are increasingly taken over by automated systems and AI-driven tools. This shift not only threatens job security for many workers but also raises concerns about the quality of work and the potential for increasing economic inequality.

The article "*The Human Cost of Digital Transformation in Financial Services*"[4] by McKinsey & Company (2021) examines the significant impact of digital transformation, especially in the financial sector. It draws conclusions very similar to those in the previous two sectors: automation and AI integration are reshaping job roles, leading to the displacement of traditional positions, particularly those involving routine tasks. The report notes that while technological advancements boost efficiency and reduce operational costs, they also pose a substantial threat to employment stability, with lower-skilled workers being the most affected.

For Professional, scientific and technical activities; administrative and support service activities, the article "*The impact of ICT professionals on job demand and employment trends*" (2023)[5], by CEDEFOP concludes that digitalization and ICT integration have significantly affected administrative jobs, leading to a shift in skill requirements. For example, demand for ICT professionals has risen, while jobs requiring lower technical skills, such as routine administrative tasks, have declined. This trend suggests a reallocation of job roles, where technological advancements push for more specialized skills, thereby impacting less-skilled workers.

On the other hand, ICT advancements can complement human labour rather than replace it, leading to higher productivity and the creation of new job opportunities. In this regard, the report "*ICTs and Jobs: Complements or Substitutes? The Effects of ICT Investment on Labour Demand by Skills and by Industry in Selected OECD Countries*"[6] by OECD researchers Vincenzo Spiezia, Michael Polder, and Giorgio Presidente (2016) analyses the effects of investments in ICT on overall labour demand, labour demand by skill level, and labour demand by industry sector in selected OECD countries between 1990 and 2012. The main goal is to determine whether ICT are complements or substitutes for human labour, assessing whether the adoption of these technologies leads to a reduction in employment (due to increased productivity) or creates new job opportunities in different sectors and emerging markets. The report shows the impact of ICT investments on labour demand, ICT can increase labour productivity, allowing the same amount of goods and services to be produced with fewer workers, potentially leading to technological unemployment. However, technological innovation also creates new job opportunities in different sectors and emerging markets. ICT investments had temporary effects on total labour demand, showing positive changes in some periods and negative in others, but there were permanent effects on labour demand by industry. Another aspect studied by OECD researchers is that ICT investments raised total labour demand in most OECD countries during the period 1990-2007 but reduced it after 2007. This reduction in labour demand was accompanied by labour polarization, with increased demand for high- and low-skilled workers at the expense of mid-skilled workers. Eventually, the effects on total labour demand and labour polarization are expected to diminish. Sectoral effects must be considered: ICT investments led to a reduction in labour demand primarily in manufacturing, business services, trade, transport, accommodation, and financial services. Sectors such as culture, recreation, construction, and public services saw an increase in labour demand due to ICT investments. The report calls for policies that foster growth in industries where ICT has positive employment effects, such as encouraging ICT adoption in these industries. It also suggests the need to support workers transitioning to new jobs, including skills development and temporary income assistance. In conclusion, the report analyses the balance between substitution effects (where ICT replaces workers) and complementarity effects (where ICT creates new job opportunities) highlighting that it depends on the economic sector and the skills required. In the production sector, the consequences are more likely to be negative; however, in the service and financial sectors, the impact depends on whether the job are traditional or emerging. The sector with the most benefits is high-tech.

Looking for articles dealing with the impact of ICT investments on employment in specific sectors, I found four articles, one for each economic activity:

- "*Technological Change and Employment in the Manufacturing Sector*"[7] by World Bank (2022)
- "*The Impact of Information and Communication Technologies on Employment in Financial Services*"[8] by Michael G. Kearney and Eoin O'Leary (2015)
- "*The Impact of ICT on Employment in the Information and Communication Sector*"[9] by Zainab A. Al-Mahmoud e Mohaisen Al-Marzouqi (2018)
- "*The Future of the Digital Workforce: Current and Future Challenges for Executive and Administrative Assistants*"[10] by Anabela Mesquita, Luciana Oliveira, and Arminda Sequeira (2019)

Even though the sectors are different, the conclusions to which all these authors arrive can be drawn together. The main result is that ICT revolutionized operations, enabling greater efficiency and speed in processes and that their implementation altered the demand for labour. On one hand, automation has reduced the need for personnel in certain traditional functions, while on the other hand, it has created new job opportunities in areas such as data management, cybersecurity, and risk analysis. All four articles emphasise the need for skills adaptation: now workers' requisites are advanced technical skills, particularly in data analytics and IT, to tackle the challenges posed by new technologies. This led to inequality in the labour market; the authors discuss how digitalization may contribute to inequalities, with some professions benefiting significantly from ICT while others may experience job losses.

In conclusion, the policy implications proposed are recommendations for policymakers, such as investing in training and reskilling the workforce to ensure workers are prepared to meet the new challenges in all sectors.

In summary, the four articles highlight that while ICT poses challenges for employment, it also offer significant opportunities for innovation and growth and institutions must address these dynamics to remain competitive in the ever-evolving landscape.

Research question

Nowadays the idea that investments in ICT increase the productivity is widespread, for example, an article by Müller and Christmann (2019), titled "*The influence of information and communication technology on productivity: A systematic review*"[11], confirms the relationship between these two variables. The authors conduct a comprehensive analysis of existing research on the impact of ICT on productivity. They review a wide range of studies to identify key themes and findings, highlighting that ICT implementation generally leads to significant productivity improvements across various sectors. The article emphasizes the importance of organizational factors, such as management practices and employee engagement, in maximizing the benefits of ICT. Overall, the systematic review provides valuable insights into the complex relationship between ICT adoption and productivity outcomes.

The main issue now is to understand if only the owners and those in high positions within the firm benefit from this improvement or if it also has a positive impact for the employees. The aim of this report is to understand the causal impact of investments in the latest technologies and employment, considering a time horizon of 20 years and focusing on four specific economic activities.

My report aims to understand the relationship between Investments in ICT equipments, software and database and annual employment, differing from the literature noted in this introduction in terms of the depth of the analysis. Thanks to the OECD database, I will try to give a more reliable overview of a current topic that is still evolving and will continue to do so in the coming years, since observations are from more countries around the world and through their evolution over the years, including different global changes. Moreover, there is a more detailed analysis in which I will not only study the four sectors chosen, but also compare them, to understand the possible implications on a case-by-case basis.

DATA

Dataset

In this study the OECD Data Explorer is the source of data; in particular, investments in ICT are provided by the STAN Database for Structural Analysis. The measure is the share of gross fixed capital formation, the economic indicator that measures the value of investments in capital goods intended to be used over a long period of time in the production of goods and services, destined to ICT equipment, software and databases and the unit of measure is millions of Euros. Annual employment is given by the dataset called “Annual employment by economic activity, domestic concept”, thanks to which we can identify the exact values for our four economic sectors:

- Manufacturing
- Information and communication
- Financial and insurance activities
- Professional, scientific and technical activities; administrative and support service activities.

Both datasets are considered from 2000 to 2019 to study their evolution over 20 years.

It is not possible to have all these data combined for all OECD nations, and the ones that have all these data available are Austria, Belgium, Canada, Czechia, Estonia, Finland, France, Italy, Japan, Latvia, Lithuania, Luxembourg, Netherlands, Norway, Slovak Republic, Slovenia, Sweden, United Kingdom, and United States.

The data for ICT investments were originally in National currency, so, to conform the observations, I chose to convert all of them in the widest spread one: Euro.

The exchange rate used to compute the changes were extrapolated by the Banca d’Italia database and referred to the end of August 2024:

Country	**National currency**	**Exchange rate**
Canada	Canadian Dollar CAD	1.5
Czechia	Czech Koruna CZK	25.27
Japan	Yen JPY	171.23
Norway	Norwegian Krone NOK	11.74
Sweden	Swedish Krone SEK	11.53
United Kingdom	British Pound GBP	0.84
United States	United State Dollar USD	1.09

Table 1

Descriptive statistics

The indices that we are going to discuss are:

- **Indices of position**: mean and median.
- **Indices of dispersion**: variance, standard deviation, and range.
- **Indices of shape**: skewness and kurtosis.

Here is a brief description of the importance of each index.

- The **arithmetic mean** represents the central value of a dataset, obtained by summing all the values of a variable and successively dividing by their counting. According to Chisini's principle, the arithmetic mean is the value that, when substituted for the individual observations, leaves the sum unchanged. It is greatly influenced by the presence of outliers, which are extreme values that are significantly distant from the rest of the data in the dataset.
- The **median** divides the ordered sequence of a variables' values into two groups of equal size. Half of the values are lower than the median, and half are higher.
- The **variance** and the **standard deviation** measure the dispersion of the data around the mean. The larger these values are, the further the data are from the mean. These two indices are highly sensitive to outliers, because extreme values are, by nature, very distant from their mean. The standard deviation is calculated as the square root of the variance.
- The **range** is obtained by subtracting the minimum value from the maximum one, giving us an idea of the variability of the data. This index is also highly influenced by outliers.
- The **skewness** provides an idea of how the data are balanced around the mean.
    - If the skewness is 0, the distribution is **perfectly symmetric**.
    - If the skewness is positive, the distribution's tail is longer on the right (**right skewness**). It is possible to achieve the same conclusion by comparing the position of the mean and the median in the boxplot: if the median is lower than the mean, there is right skewness.
    - If the skewness is negative, (**left skewness**). Similarly, the same conclusion can be drawn if the mean is lower than the median.
- The **kurtosis** measures the distribution's shape, indicating how sharp the peaks are and how long the tails are compared to a normal distribution.
    - When the kurtosis is positive, the distribution is called **leptokurtic** and has higher peaks and longer tails, with extreme values occurring more frequently than in a normal distribution.
    - A negative kurtosis (**platykurtic**) means that the distribution is characterised by lower peaks and shorter tails, with fewer extreme values than a normal distribution.
    - The distribution resembles a normal one when the kurtosis is around 0 (**mesokurtic**).

This study has three levels of analysis: Country, Year and Economic activity, so the descriptive statistics will follow this subdivision.

- ## Data grouped by Country
    - ### Investments in ICT equipment, software and database

INVESTMENTS IN ICT/ GDP PER COUNTRY							
country	mean	median	var	sd	range	skew	kurtosis
Austria	9.29%	6.02%	0.0063	0.0797	39.36%	1.5335	2.2825
Belgium	7.21%	5.00%	0.0027	0.0522	18.58%	1.0222	-0.2527
Canada	5.43%	3.74%	0.0021	0.0458	21.94%	1.6458	2.6052
Czechia	10.35%	7.17%	0.0048	0.0691	22.87%	0.4795	-1.3064
Estonia	5.39%	3.59%	0.0016	0.0405	13.33%	1.1822	-0.0393
Finland	5.22%	4.57%	0.0008	0.0289	11.87%	0.3608	-1.0797
France	8.92%	7.73%	0.0017	0.0416	15.54%	0.8400	-0.3430
Italy	6.56%	2.87%	0.0048	0.0696	23.85%	1.3106	0.0300
Japan	7.86%	5.65%	0.0028	0.0531	16.69%	0.7409	-1.0347
Latvia	4.59%	2.89%	0.0019	0.0435	13.74%	0.8816	-0.6325
Lithuania	6.65%	5.03%	0.0028	0.0528	19.23%	1.2398	0.4929
Luxembourg	3.42%	1.66%	0.0010	0.0323	12.27%	1.0782	-0.2370
Netherlands	5.19%	4.05%	0.0008	0.0285	15.93%	1.8184	4.4683
Norway	4.53%	2.84%	0.0014	0.0379	16.53%	1.4278	1.3969
Slovak Republic	7.50%	4.02%	0.0080	0.0897	38.39%	1.6219	1.9750
Slovenia	7.66%	4.69%	0.0055	0.0743	34.03%	1.1813	0.9500
Sweden	11.23%	8.52%	0.0059	0.0770	29.90%	1.0245	0.0660
United Kingdom	5.62%	4.06%	0.0018	0.0423	20.14%	1.7126	2.8850
United States	7.28%	5.80%	0.0026	0.0509	22.83%	1.1261	0.5862

Table 2
Unit of measure: millions of Euros

Table 2 displays the indices for data grouped by country. To make them comparable, the Investments in ICT are divided by the GDP of the sector in each year and in each country, so that the values represent the percentage of GDP invested in ICT.

There are big differences among the means: the highest value is Sweden, with 11.23% of GDP invested in ICT, followed by the 10.35% in Czechia. The large global economics have an average spending of around 8% and another European country that invest more than them is Austria (9.29%). The US percentage is 7.29%, while, among the European countries, the nation that invests the most is France with 8.92%, ahead of Italy (6.56%) and the UK (5.62%). Globally, another important figure is Japan, with an average spending of 7.86%. The nations with the lowest percentages are Latvia (4.59%), Norway (4.53%) and Luxembourg (3.42%).

All the medians are lower than the corresponding means, so it is likely that the distributions have right skewness. These assumptions need to be compared with skewness values, which confirm the earlier hypothesis of distributions with longer tail on the right.

The variance among the countries' data would have been extremely high without the introduction of the ratio between ICT investments and GDP, for a few reasons. By dealing with more dimensions, I can stratify the data, making the variability within each group clearer. For example, it is possible that a specific sector in a specific country has a different variance from that of another sector in the same country. Similarly, including years makes aware of the different variability over time due to economic, political, or social factors. The two highest values are Austria and Sweden, two of the three countries with the highest average spending, with values of 0.0063 and 0.0059. The lowest value is a tie between Netherlands and Finland (0.0008).

The only countries where the range is above 30% are Slovenia (34.03%), the Slovak Republic (38.39%), and Austria (39.36%), while the smallest difference between maximum and minimum values is 11.97%, belonging to Finland.

The kurtosis index shows that six countries are platykurtic (Belgium, Czechia, Finland, France, Japan and Luxembourg) because they have negative values, while the remaining countries, with positive values, have higher peaks and longer tails than a normal distribution. Three countries can be seen as mesokurtic, since their values are only slightly different from zero: Sweden (0.0660), Estonia (0.0393), and Italy (0.0300).

- Employment

EMPLOYMENT RATE PER COUNTRY							
country	mean	median	var	sd	range	skew	kurtosis
Austria	1.50%	1.23%	0.0007	0.0269	13.53%	0.7091	0.2891
Belgium	0.81%	0.24%	0.0008	0.0282	13.71%	0.5039	-0.2518
Canada	1.09%	1.21%	0.0006	0.0253	14.44%	-0.8581	1.7320
Czechia	1.37%	1.47%	0.0009	0.0305	15.93%	-0.4781	0.8274
Estonia	3.33%	2.65%	0.0124	0.1115	58.78%	0.5410	0.6264
Finland	1.43%	0.99%	0.0012	0.0341	21.20%	0.1522	1.1334
France	0.78%	0.70%	0.0005	0.0231	12.83%	0.2126	0.9597
Italy	0.77%	0.54%	0.0006	0.0237	13.87%	0.6568	1.1801
Japan	0.71%	0.81%	0.0008	0.0279	13.29%	-0.0925	-0.2618
Latvia	1.32%	1.03%	0.0056	0.0750	47.69%	-0.0526	2.0253
Lithuania	2.43%	2.11%	0.0101	0.1004	60.08%	0.4124	0.9243
Luxembourg	3.38%	2.70%	0.0013	0.0362	15.78%	0.6715	-0.2419
Netherlands	0.31%	-0.06%	0.0009	0.0295	14.69%	0.3983	-0.3862
Norway	0.71%	0.73%	0.0012	0.0350	21.16%	0.5631	1.3208
Slovak Republic	2.02%	1.86%	0.0017	0.0406	21.50%	-0.3124	0.7676
Slovenia	1.85%	1.84%	0.0018	0.0420	32.88%	1.4850	7.4364
Sweden	1.09%	1.45%	0.0012	0.0350	23.98%	0.4436	2.1188
United Kingdom	0.49%	0.90%	0.0007	0.0270	12.09%	-0.4584	-0.2934
United States	0.29%	0.97%	0.0009	0.0304	15.68%	-1.2039	1.6247

Table 3
Unit of measure: Persons, Thousands

Table 3 shows the differences in countries for the employment; in particular, to have more homogenous observations, in this case, I considered the employment rate.

In Luxembourg, there was the most average employment growth from 2000 to 2019 in the four sectors considered: 3.38%. The second was Estonia (3.33%), and the only other two countries that exceeded the threshold of 2% were Slovak Republic (2.02%) and Lithuania (2.43%). The lowest average growth was in the United States, with a value of 0.29%.

There are seven medians bigger than the mean in *Table 3*: Canada, Czechia, Japan, Norway, Sweden, the United Kingdom, and the United States, suggesting possible left skewness in these countries. Checking the relative index, this assumption is correct for Canada, Czechia, Japan, the United Kingdom, and the United States but incorrect for Norway and Sweden. This discrepancy can be attributed to the presence of a limited number of extremely high values, which can influence the mean more than the median, lowering it. This could lead to an inversion of the median and mean values, even though the skewness remains positive due to the length of the right tail. Reasoning inversely, it's possible to understand why Latvia and Slovak Republic have a mean greater than the median but exhibit a left skewness.

There are two countries that have a variance of more than 0.01: the highest one is Estonia (0.0124), followed by Lithuania (0.0101). On the opposite side, there are a lot of countries under 0.0010, with the lowest variance observed in France (0.0005).

Lithuania has the widest range (60.08%), followed by Estonia (58.78%). Many countries have a range between 15% and 10%, and the lowest of them is United Kingdom (12.09%).

The platykurtic distributions are Belgium, Japan, Luxembourg, Netherlands, and United Kingdom. All the other countries have a positive kurtosis, indicating a leptokurtic distribution.

- Data grouped by Year

o Investments in ICT equipment, software and database

INVESTMENTS IN ICT/GDP PER YEAR										
year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
mean	8.15%	7.79%	7.04%	6.65%	6.86%	6.69%	6.71%	6.67%	6.80%	6.34%
median	5.69%	4.96%	4.76%	4.87%	4.72%	4.98%	4.67%	4.79%	5.01%	4.59%
var	0.0062	0.0049	0.0041	0.0028	0.0036	0.0031	0.0030	0.0031	0.0030	0.0026
sd	0.0788	0.0701	0.0640	0.0529	0.0601	0.0556	0.0549	0.0560	0.0549	0.0506
range	40.60%	32.34%	27.32%	22.59%	35.31%	28.67%	23.25%	28.28%	23.70%	20.06%
skew	1.8884	1.4374	1.5157	1.2296	1.9450	1.4542	1.1514	1.4584	1.1389	1.1506
kurtosis	3.8942	1.4521	1.5516	0.8714	5.4332	2.3298	0.4435	2.1943	0.6609	0.3892
year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
mean	6.51%	6.44%	6.34%	6.58%	6.49%	6.87%	6.82%	7.14%	7.13%	6.69%
median	4.67%	4.54%	4.28%	4.32%	4.15%	4.69%	4.78%	4.69%	4.85%	4.73%
var	0.0037	0.0030	0.0029	0.0032	0.0034	0.0038	0.0034	0.0039	0.0040	0.0030
sd	0.0610	0.0547	0.0539	0.0562	0.0579	0.0613	0.0586	0.0625	0.0632	0.0551
range	38.00%	25.13%	25.70%	24.72%	25.10%	26.54%	24.06%	32.44%	31.44%	25.72%
skew	2.4362	1.2987	1.3963	1.4080	1.5107	1.4760	1.2062	1.5859	1.5204	1.2844
kurtosis	8.5279	1.0176	1.4727	1.2713	1.4977	1.6865	0.5124	2.7129	2.2966	1.3086

Table 4
Unit of measure: millions of Euros

In *Table 4* and *Table 5*, it is possible to observe the temporal development of the data.

The mean ratio between ICT investments and GDP reached its maximum value at the beginning of the selected interval: in 2000 it was 8.15%. In 2001 and 2002 it dropped to 7.79% and 7.04%, falling below 7% in 2003. Only in 2017 the percentage exceeded the 7% threshold again, but by 2019 it had returned to 6.69%.

In every year, the median is lower than the corresponding mean, suggesting positive skewness. This assumption can be confirmed by looking at the skewness values: they are all positives, indicating that the distribution has a longer tail on the right side.

The variance followed the mean's trend: when the mean increased, the variance increases, and vice versa. Indeed, the largest variance was in 2000 (0.0062).

The range of observations varied over the years, alternating between values close to 40% and values around 25%. The widest range was in 2000, but it also reached a high value in 2010 (38%) and in 2004 (35.31%). Among other years, the smallest range was in 2009 (20.06%).

Although some kurtosis indices are close to 0, every value is positive, so the distribution is leptokurtic, with higher peaks, longer tails, and more frequent extreme values than in a normal distribution.

o Employment

To better compare the data and to avoid the influence of extreme values, in *Table 5*, as in *Table 3*, I'm going to study the employment rate, this time between 2000 and 2019.

The highest employment rate was in 2007, right before the 2008 crisis, which showed its effects in the 2009 and 2010 values, as not only they are negative, but they are also the lowest of the time series (-2.98% and -1.41%). The only other year with a negative employment rate was 2003, but -0.16% is

EMPLOYMENT RATE PER YEAR										
year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
mean	3.30%	1.02%	0.86%	-0.16%	0.46%	1.76%	2.36%	4.26%	2.63%	-2.98%
median	2.21%	1.06%	0.42%	-0.47%	-0.30%	1.39%	2.02%	2.96%	2.18%	-2.31%
var	0.0044	0.0026	0.0040	0.0022	0.0028	0.0031	0.0016	0.0045	0.0019	0.0028
sd	0.0662	0.0512	0.0633	0.0472	0.0532	0.0561	0.0394	0.0674	0.0438	0.0530
range	46.76%	35.20%	41.93%	34.55%	39.82%	39.79%	21.86%	44.53%	29.06%	32.01%
skew	2.0822	-1.5921	1.6949	1.7129	1.8011	-0.4820	0.2587	2.3374	0.4845	-0.8213
kurtosis	9.1237	6.4129	5.6450	8.1572	7.9937	4.7642	0.5265	7.3971	2.3317	1.6288
year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
mean	-1.41%	2.34%	1.48%	0.50%	0.58%	1.88%	2.11%	2.23%	2.00%	1.80%
median	-0.88%	1.56%	0.90%	0.10%	0.40%	1.51%	2.18%	2.29%	1.81%	1.31%
var	0.0014	0.0025	0.0012	0.0011	0.0015	0.0014	0.0009	0.0010	0.0010	0.0007
sd	0.0379	0.0502	0.0340	0.0335	0.0392	0.0375	0.0300	0.0323	0.0319	0.0258
range	22.38%	44.01%	24.01%	23.83%	31.85%	22.54%	20.36%	18.93%	23.15%	17.15%
skew	-1.4487	2.5999	2.5549	0.3056	-1.3131	2.2408	1.2307	1.1704	0.2890	1.9166
kurtosis	3.9172	14.0782	10.2012	3.9165	8.6747	7.1575	4.2636	2.7589	3.3355	6.8452

Table 5
Unit of measure: Persons, Thousands

much closer to 0%. In all other years, the value approached 0% at times, but from 2015 they remained stable above 1.8%, and even exceeded 2% in 2016, 2017, and 2018.

In this time series, it is more common for the median to be lower than the mean, but this did not happen in all the years analysed: 2001, 2009, 2010, 2016, and 2017 have a median higher than the mean, so only for these years I assume that there is a left skewness. However, as in the previous analysis, the skewness values do not confirm my assumptions: 2016 and 2017 have positive values and, on the other hand, 2005 and 2014 have negative ones, indicating that in these two years the distribution's tail longer on the left. For all the years with positive values, the tail is longer on the right.

The largest values of the variance are in 2017 (0.0045) and 2000 (0.0044), and as the years passed, the variance decreased, ending in 2019 with a value of 0.0007.

The ranges in this time series are not stable, changing considerably from year to year, but from 2014 onwards, the range decreased, eventually reaching a difference between the maximum and minimum values of 17% to 24%. Only in 2006, 2010, and 2012 the ranges were so low.

In conclusion, the kurtosis indices assume positive values throughout the years, so once again the distribution is leptokurtic.

- Data grouped by Economic activity

o Investments in ICT equipment, software and database

INVESTMENTS IN ICT/GDP PER ECONOMIC ACTIVITY							
economic activity	mean	median	var	sd	range	skew	kurtosis
Financial and insurance activity	6.65%	6.22%	0.00	0.03	16.93%	0.69	0.14
Information and communication	14.63%	14.01%	0.00	0.06	35.79%	1.03	1.74
Manufacturing	2.56%	2.23%	0.00	0.02	8.30%	1.49	2.99
Professional, scientific and technical activities; administrative and support service activities	3.51%	3.47%	0.00	0.01	8.80%	0.20	-0.11

Table 6
Unit of measure: millions of Euros

Thanks to *Table 6* and, later, *Table 7*, it is possible to compare the four economic activities selected. Once again, the first table shows the proportion of GDP invested in ICT, and the second one displays the employment rate.

Table 6 focuses on investments in ICT, and among all the countries selected from 2000 to 2019, the sector with the highest mean spending is information and communication (14.63%), which is almost the double of the second highest one. The other three activities have closer averages, but the lowest is manufacturing, with 2.56% invested.

The medians follow the same order as the means: the largest sector remains information and communication (14.01%), and manufacturing remains the lowest (2.23%). Every median is lower than the average, suggesting right skewness, which skewness values confirm.

The variances among sectors are so low that, when rounded to two decimals, they do not differ from 0. The standard deviations help to understand the variability between sectors, and, in this case, the lowest variability is found in professional, scientific and technical activities; administrative and support service activities.

The range follows the trend of the previous indices: information and communication sector has the largest difference between the maximum and minimum values (35.79%), while the manufacturing sector has the smallest range (8.30%).

The first three sectors analysed have positive kurtosis indices, indicating that their distribution is leptokurtic. However, the last sector has lower peaks and shorter tails, with fewer extreme values than a normal distribution.

- Employment

EMPLOYMENT RATE PER ECONOMIC ACTIVITY							
economic activity	mean	median	var	sd	range	skew	kurtosis
Financial and insurance activity	0.52%	0.13%	0.00	0.05	60.08%	1.36	14.06
Information and communication	2.53%	2.23%	0.00	0.06	59.61%	1.19	8.19
Manufacturing	-0.72%	-0.35%	0.00	0.03	34.95%	-1.53	8.52
Professional, scientific and technical activities; administrative and support service activities	3.08%	2.76%	0.00	0.04	38.94%	0.99	5.41

Table 7
Unit of measure: Persons, Thousands

Table 7 displays the employment rate of the four economic activities across the nineteen countries for the years analysed.

The mean employment shows a clear distinction between sectors: information and communication and professional, scientific and technical activities; administrative and support service activities have values above 0, while financial and insurance activities, though positive, is closer to 0. The manufacturing sector is the only one where the average employment rate is negative (-0.72%).

For the positive means, the medians are lower, while for manufacturing it is the median higher. Every assumption made according to these comparisons is confirmed by the skewness values. Only the manufacturing sector shows left skewness.

The variances face the same issue as in *Table 6*, but using standard deviations, I can rank them by variability. The information and communication sector has the highest (0.06), and the manufacturing sector the lowest (0.03).

The final index is kurtosis, and since all values are positive, it can be concluded that the four sectors have a leptokurtic distribution, with higher peaks and longer tails, and more extreme values than a normal distribution.

REGRESSION ANALYSIS

Assumptions

There are five assumptions that must be satisfied to study the relation between the variables of interest using Ordinary Least Squares (OLS).

- **Linearity**

There is an assumed linear relationship between the independent variable and the dependent variable. If the data are heavily skewed, the relationship may appear non-linear. Logarithmic transformations can help linearize relationships that are not obvious in the original data. If the relationship is non-linear, OLS will produce biased estimates of the coefficients because the model cannot capture the true relationship between the variables.

- **Independence of errors**

The errors of the model must be independent of each other. If the original variables have outliers or are strongly skewed, the presence of these outliers can influence the independence of errors. Logarithmic transformations can reduce outliers, thereby improving the independence of errors. When errors are not independent, the standard error estimates may be underestimated or overestimated, making significance tests unreliable.

- **Normality of errors**

The errors of the model should follow a normal distribution. The distribution of errors in the original data can be heavily skewed, making it difficult to satisfy this assumption. Logarithmic transformations make the distributions more symmetric and closer to normality. It is important for the validity of statistical inferences that rely on the normal distribution of the residuals. If the errors are not normally distributed, confidence intervals and hypothesis tests may not be accurate, especially in small samples. However, with large samples, the central limit theorem can mitigate this issue.

- **Homoskedasticity**

Homoscedasticity ensures that the variance of the errors is constant across all observations. While OLS estimators remain unbiased in the presence of heteroscedasticity, they are no longer efficient. When homoscedasticity is violated (heteroscedasticity), the standard error estimates of OLS become unreliable, leading to inaccurate significance tests. Logarithmic transformations stabilise the variance.

- **No multicollinearity**

The independent and dependent variables should not be highly correlated with each other. OLS estimators can still be calculated in the presence of multicollinearity, but the estimates of the coefficients become unstable and may have very high variances. If there is high multicollinearity, it becomes difficult to determine the separate effect of each predictor. The estimates may become very sensitive to changes in the data, making it hard to interpret the coefficients.

In conclusion, to ensure that OLS estimators are unbiased, efficient, and consistent, it is essential that these assumptions are met. If one or more assumptions are violated, OLS estimates may still be unbiased but not efficient, or they may be completely inadequate to represent the data.

Before starting with the regression analysis, it is necessary to determine whether data can or cannot satisfy these assumptions. In particular, the main topic is to decide whether use logarithmic transformations or not. To give a comprehensive overview, *Figure 1* and *Figure 2* display the boxplot and the distribution plot, respectively.

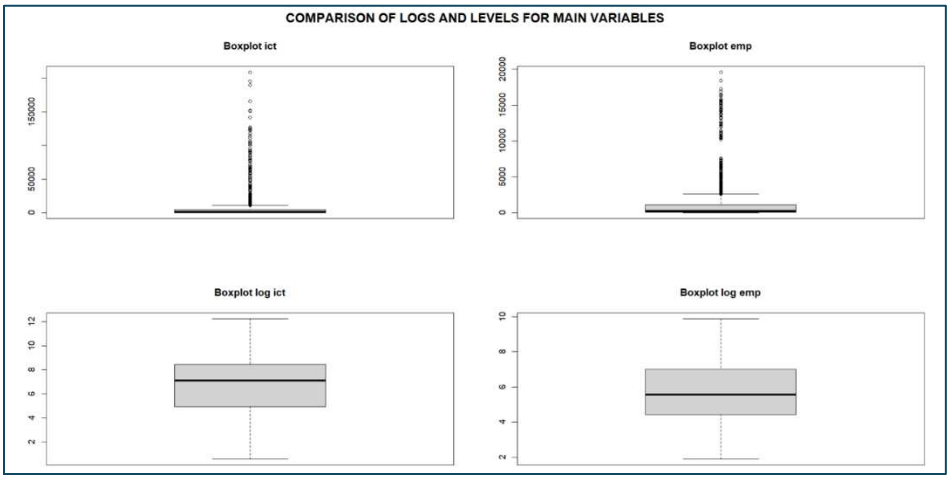

Figure 1

Because of the large amount of data and their different nature, the logarithmic form seems more suitable to face a regression model; in fact, both *log ict* and *log emp* eliminate all the outliers.

Figure 2 shows that, thanks to the logarithmic transformation, there is a reduction in the distribution's asymmetry.

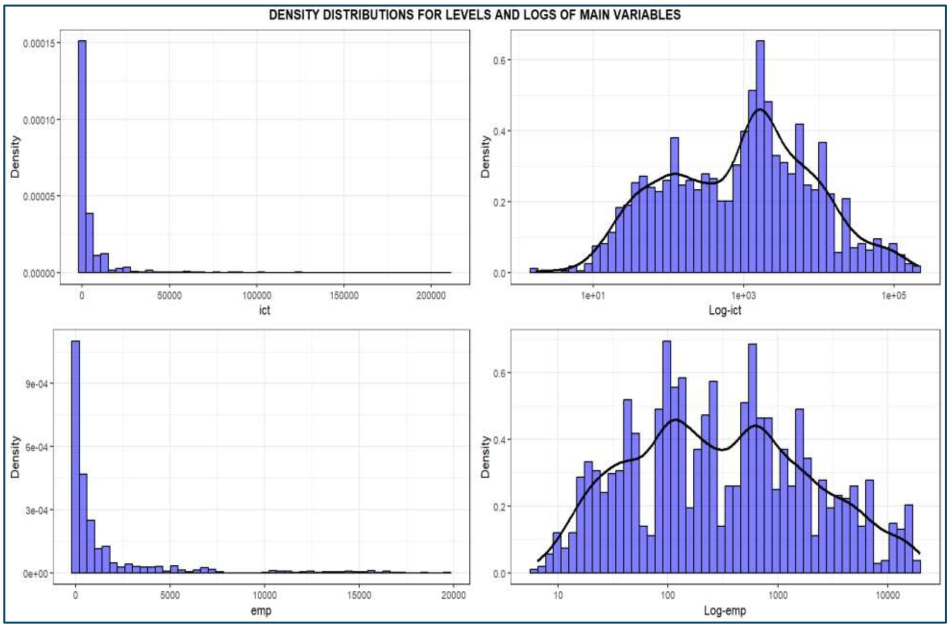

Figure 2

To determine whether investments in ICT and annual employment satisfy this condition, I run an Rstudio script for both values with and without logarithmic transformation.

Without logarithmic transformation, the correlation results in 0.63, while log-transforming them, it increases to 0.82. This means that the two variables move together, but it may be because of a third element, so only computing the regression model we can clarify their relationship.

When using original variables that do not meet the assumptions, the regression model can produce distorted results, incorrect standard errors, and unreliable statistical inferences. Therefore, transforming variables using logarithms makes the data more compliant with regression assumptions, thereby improving the accuracy and validity of the model.

Regression models

- Model 1 and Model 2: Levels vs Logs

Basing on the previous considerations, a log-log model should end up in a better model to understand the relationship between ICT investments and Annual employment.

Now, thanks to R, I will compare Level-Level model

$$(1) \quad EMP_{i,j,t} = \beta_0 + \beta_1 * ICT_{i,j,t} + u_{i,j,t}$$

and the Log-Log model

$$(2) \quad \log(EMP)_{i,j,t} = \beta_0 + \beta_1 * \log(ICT)_{i,j,t} + u_{i,j,t}$$

Where:

- i is the subscript representing the 19 countries selected.
- j is the subscript for the four economic sectors.
- t stands for the time dimension from 2000 to 2019.
- $EMP_{i,j,t}$ is the dependent variable Y.
- $ICT_{i,j,t}$ is the independent variable X.
- $\log(EMP)_{i,j,t}$ is the logarithmic form of the previous dependent variable annual employment.
- $\log(ICT)_{i,j,t}$ is the logarithmic form of the previous independent variable Investments in ICT equipment, software and database.
- β_0 is the intercept, so the value that the dependent variable assumes when the independent one is equal to 0.
- β_1 is the slope coefficient. In the Level-Level model it indicates the change in the dependent variable for a unit change in the independent one, in the Log-Log one it is the percentage change of Y for a 1% increase in X.
- $u_{i,j,t}$ is the error term, so the difference between observed and predicted values.

The main difference between *Models 1* and *2* is the interpretation of the coefficients.

Model 1 is the simplest model, because it creates a model based only on the data as they appear in the dataset. The results are that for a model with ICT investments equal to 0, the annual employment will be of 749,668 persons, while for an increase of 1 million euros in spending, the employees will increase of 93. Both the parameters are statistically significant, since $t_{\beta 0} = \frac{749.668}{61.169} = 12.26$ and $t_{\beta 1} = \frac{0.093}{0.003} = 31$ are higher than the critical value for $\alpha = 1\%$ (2.58).

LEVELS vs LOGS		
	Level-Level (1)	Log-Log (2)
(Intercept)	749.668***	1.069***
	(61.159)	(0.085)
ict	0.093***	
	(0.003)	
log_ict		0.684***
		(0.012)
Num.Obs.	1520	1520
R2	0.401	0.687
R2 Adj.	0.400	0.687
* p < 0.1, ** p < 0.05, *** p < 0.01		

Summary 1

This probably isn't the best model to study these data, because it doesn't consider the possible correlations among the same category of data, but it's important to comment on it anyway to have a starting point.

Model 2 (Log-Log) has $\beta_0 = 1.069$, meaning that when investments in ICT is equal to 1, so equal to the unit of measurement 1,000,000 €), the Annual employment has a value of 2912 workers.

Being a log-log model, the consequence of having $ICT_{i,j,t} = 1$ is the following:

$$\log(EMP)_{i,j,t} = 1.069 + \ 0.684 \log(1)_{i,j,t} = 1.069$$

$$EMP_{i,j,t} = e^{1.069} = 2.912$$

$\beta_1 = 0.684$ shows a positive relationship, in particular for an increase of 1% in the investments in ICT the annual employment grows of 0.684%. Now we must verify the significance of the parameters thanks to the formula used also for *Model 1*:

$$t = \frac{value}{stadard\ error}$$

For β_1 the significance level is $\frac{0.684}{0.012} = 57$ and for β_0 it's s $\frac{1.069}{0.085} = 12.58$. Since both are bigger than 2.58, the two parameters are statistically significant even at 1% level.

Concluding the dissertation on *Models 1* and *2*, we found out a better R^2 value for the log-log model, even though it is not primary for this report. Thus, the 68.7% of the variability in annual employment is explained by investments in ICT.

Also *Model 2* doesn't solve the problem of correlation within panel data, but the values just commented, along with the boxplots and the density plots, suggest that the initial model of the regression analysis must be the use log-transformed.

The literature also confirms the usefulness of using logarithms in the study of ICT investments. In particular, "*Information technology and the U.S. productivity Revival: What Do the Industry Data Say*?"[12] by Kevin J. Stiroh (2002) analysed the role of ICT in the recovery of productivity of USA in the ‘90s. Stiroh examines industry level data to understand how investments in ITC helped productivity’s growth.

The author prefers log-log regression models because of some advantages:

- **Elasticity interpretation**: coefficients in log-log models represent elasticities, making them easier to interpret as percentage changes.
- **Variance stabilization**: log transformations reduce the effect of outliers and heteroscedasticity, improving the robustness of the estimates.
- **Comparability**: logarithms allow a better comparability across industries of different scales and over time.

Stiroh concludes that using logs improves both the precision and interpretability of his productivity models.

- Model 3: Entity fixed effects

Model 2 isn’t the most suitable for my data, mainly for two reasons: the first is internal correlation, because in a panel database there is an intrinsic correlation between observations of the same country over time. These correlations violate the assumption of the independence of observations required for a simple linear model. The second reason is called “Omitted Variable Bias” (OVB), a systematic bias of the OLS for the causal effect of X on Y. To be called omitted variable, two conditions need to be met: a regressor X must be correlated with the omitted variable, and the omitted variable is a determinant of Y.

	Log-Log (2)	Entity FE (3)
(Intercept)	1.069***	
	(0.085)	
log_ict	0.684***	−0.079
	(0.012)	(0.091)
Num.Obs.	1520	1520
R2	0.687	0.804
R2 Adj.	0.687	0.801
R2 Within		0.004
R2 Within Adj.		0.003

* p < 0.1, ** p < 0.05, *** p < 0.01

Summary 2

Adding country fixed effects α_i means controlling for omitted variables in panel data when the omitted variables vary across entities but do not change over time. For example, a country’s size, which impacts employment distribution and infrastructure needs, and economic policies with structural differences in fiscal policies. Another example is the average level of economic development (GDP per capita), which influences ICT investment and labour demand in a stable way.

$$(3) \quad \log(EMP)_{i,j,t} = \beta_1 * \log(ICT)_{i,j,t} + \alpha_i + u_{i,j,t}$$

By looking at *Summary 2*, it's easily understandable that, thanks to fixed effects the results are totally different from *Model 2*. The coefficient β_1 changes the direction, so now for an increase of 1% in investments in ICT equipment, software and database, annual employment drops by 0.79%. This isn't the only important change; also, checking the level of significance, we find a substantial difference from *Model 2:* $t_3 = \frac{0.079}{0.091} = 0.86$. In this case, the parameter is no longer statistically significant, not even for $\alpha = 10\%$ $(0.86 < 1.64)$. The only element that improved is the R^2, which in *Model 3* is 80.4%.

- Model 4: Entity and time fixed effects

Continuing the previous reasoning, the country dimension isn't the only one for which fixed effects can be added, since there may be factors that vary over time but are common to all states and economic activities in this period. A few examples are the 2008 crisis, some changes in monetary or fiscal policies, such as central bank interest rates, and new international regulations or decisions by supranational organizations that affect all involved states simultaneously. In the end, some global social or demographic trends can shift in consumer behaviour or demographic patterns, such as the increased use of digital technologies.

$$(4) \quad \log(EMP)_{i,j,t} = \beta_1 * \log(ICT)_{i,j,t} + \alpha_i + \lambda_t + u_{i,j,t}$$

	Log-Log (2)	**Entity FE (3)**	**Entity + time FE (4)**
(Intercept)	1.069***		
	(0.085)		
log_ict	0.684***	−0.079	−0.131
	(0.012)	(0.091)	(0.099)
Num.Obs.	1520	1520	1520
R2	0.687	0.804	0.806
R2 Adj.	0.687	0.801	0.801
R2 Within		0.004	0.009
R2 Within Adj.		0.003	0.008
* p < 0.1, ** p < 0.05, *** p < 0.01			

Summary 3

The results of *Model 4* are slightly different, but they retain the main characteristics of *Model 3*: β_1 is negative and not statistically significant. Now, annual employment decreases by 0.131% for a 1% increase in investments in ICT and $t_4 = \frac{0.131}{0.099} = 1.32$.

Fixed effects captured omitted variables that previously influenced both X and Y, giving a false significance. By adding fixed effects, the relationship between investments in ICT and annual employment collapsed, suggesting that the initial correlation was due to omitted variables which weren't considered in *Model 2*.

- Model 5: Entity, time and sector fixed effects

The last model, *Model 5*, corrects for the third dimension of this analysis: the economic activity. Following the same reasoning behind the first two dimensions, it is possible to think of aspects that vary within the specific economic activity but not overt time or across states. These effects reflect industry-specific factors that consistently influence employment or ICT investments. Examples of such fixed effects include technological differences, industry productivity, affecting labour demand or ICT investment, and activity-specific regulations that do not change frequently or significantly over time.

$$(5) \quad \log(EMP)_{i,j,t} = \beta_1 * \log(ICT)_{i,j,t} + \alpha_i + \lambda_t + \gamma_j + u_{i,j,t}$$

Summary 4 sums up the results obtained by gradually implementing the three fixed effects.

FIXED EFFECTS MODELS				
	Log-Log (2)	Entity FE (3)	Entity + time FE (4)	Entity + time + sector FE (5)
(Intercept)	1.069***			
	(0.085)			
log_ict	0.684***	−0.079	−0.131	0.337***
	(0.012)	(0.091)	(0.099)	(0.092)
Num.Obs.	1520	1520	1520	1520
R2	0.687	0.804	0.806	0.974
R2 Adj.	0.687	0.801	0.801	0.974
R2 Within		0.004	0.009	0.225
R2 Within Adj.		0.003	0.008	0.225
* p < 0.1, ** p < 0.05, *** p < 0.01				

Summary 4

After adding the last dimension in the fixed effects, the results seem more similar to those of *Model 2*. Not only the parameter regain statistical significance at 1% level $\left(t = \frac{0.337}{0.092} = 3.66 > 2.58\right)$, but the relationship between investments in ICT and annual employment is again positive. In this model, a 1% increase in ICT investments corresponds to an increasing of 0.337% in annual employment. In this case, the power of the estimation is more theoretically supported. The fact that β_1 becomes significant again after adding fixed effects for the economic sector indicates that differences between economic sectors have a big impact on the relationship between X and Y. The effect of X was hidden or distorted without this control, and the inclusion of the third dimension allowed for the detection of a real and significant relationship between the independent variable (investments in ICT) and the dependent variable (annual employment), suggesting that the sectoral context plays a crucial role in this relationship.

In *Model 5*, even the R^2 is the highest registered with an extremely high value of 97.4% of the variability explained by the independent variable.

Concluding the study of this three dimensional models, *Model 5* is the most suitable for all the reasons just listed. It's important to underline that this does not mean that this model is the best one possible to study this relationship, as there may be some other omitted variables that aren't considered.The filters from which I found the initial data are quite strict, and, because of this, it's difficult to find observed data to add to the model, for example the educational level of the workforce and tax incentives.

Submodels on Economic activities

The literature mentioned in the Introduction underscored the need to study the impact of investments in ICT equipment, software and database on annual employment, focusing on one economic activity at a time. So, this will be the aim of the following submodels. The steps will retrace the ones that led me from an initial and inadequate *Model 2* to the more suitable and supported *Model 5*.

- Manufacturing

MANUFACTURING			
	Log-Log (2_A)	Entity FE (3_A)	Entity + time FE (4_A)
(Intercept)	2.381***		
	(0.095)		
log_ict	0.631***	−0.041	0.043*
	(0.013)	(0.026)	(0.024)
Num.Obs.	380	380	380
R2	0.853	0.997	0.999
R2 Adj.	0.853	0.997	0.999
R2 Within		0.018	0.044
R2 Within Adj.		0.016	0.041

* p < 0.1, ** p < 0.05, *** p < 0.01

Summary 5

As for the regression model with three dimensions, the first step is to compute the log-log model, but this time only for data belonging to manufacturing. The intercept $\beta_1 = 0.631$ implies that for a 1% increase in ICT investments in manufacturing, annual employment increases by 0.631%. The parameter is statistically significant for $\alpha = 1\%$, since $t = \frac{0.631}{0.013} = 48.54 > 2.58$.

The next step to improve *Model 2_A* is adding the entity fixed effects. After this, the results change drastically, as displayed in *Model 3_A*: not only the coefficient inverted the sense of the relationship, but also it loses significance. Now, for a 1 percentage point growth in X, Y decreases of 0.041% and $t = \frac{0.041}{0.026} = 1.57$ isn't even bigger than 1.64, the critical value for $\alpha = 10\%$.

The situation improves after including the year fixed effects, thanks to them there is again a positive relationship between the dependent and independent variables, with a 0.043% growth in annual employment for a 1% increase in ICT investments in the manufacturing sector. The parameter becomes statistically significant for $\alpha = 10\%$, since $t = \frac{0.043}{0.024} = 1.792$ is greater than 1.648, but lower than both 1.96 and 2.58.

Even the R^2 shows a gradual improvement in the model, ending up with an extremely high value of 0.99.

- Information and communication

INFORMATION AND COMMUNICATION			
	Log-Log (2_B)	Entity FE (3_B)	Entity+time FE (4_B)
(Intercept)	−0.811***		
	(0.077)		
log_ict	0.791***	0.276***	0.089
	(0.010)	(0.045)	(0.060)
Num.Obs.	380	380	380
R2	0.942	0.995	0.997
R2 Adj.	0.942	0.995	0.997
R2 Within		0.417	0.060
R2 Within Adj.		0.415	0.057
* p < 0.1, ** p < 0.05, *** p < 0.01			

Summary 6

The initial condition in *Summary 6* is shown by *Model 2_B*, where for a unitary value of ICT investments, the log transformation of annual employment is -0.811.

$$\log(EMP) = -0.811 + 0.791 * \log(1)$$

$$\log(EMP) = -0.811$$

$$EMP = 0.444$$

On the other hand, $\beta_1 = 0.791$ means that for an increase of 1% in ICT investments, the annual employment increase of 0.791% and, with a $t = \frac{0.791}{0.010} = 79.1$, the parameter is statistically significant for an $\alpha = 1\%$.

In *Model 2_B*, adding entity fixed effects, the coefficient maintains its significance: $t = \frac{0.276}{0.045} = 6.13 > 2.58$. Now a growth of one percentage point in X is followed by a growth of Y of 0.276%, so the variables remain positively related. This implies that the relationship between β_1 and the dependent variable persists even after controlling for unobservable differences across entities, strengthening the validity of the parameter and excluding the risk that it was influenced only by heterogeneity between entities.

Since in *Model 4_B* β_1 loses its statistical significance when time fixed effects are added, the relationship between the independent variable and the dependent variable is influenced by temporal factors. This may indicate that some of the observed effect of the coefficient was driven by common time-specific influences, such as macroeconomic conditions, policies, or global trends affecting all entities similarly.

Once again, the R^2 increases as the model evolves, reaching in *Model 4_B* a value of 99.7% of the variability explained.

- Financial and insurance activities

FINANCIAL AND INSURANCE ACTIVITY			
	Log-Log (2_C)	Entity FE (3_C)	Entity+time FE (4_C)
(Intercept)	−0.331***		
	(0.086)		
log_ict	0.800***	0.052	0.013
	(0.012)	(0.041)	(0.047)
Num.Obs.	380	380	380
R2	0.916	0.998	0.998
R2 Adj.	0.916	0.998	0.998
R2 Within		0.044	0.002
R2 Within Adj.		0.042	−0.001
* p < 0.1, ** p < 0.05, *** p < 0.01			

Summary 7

The results displayed in *Summary 7* resemble the ones in *Summary 6;* therefore, according to the chosen models, the effects in information and communication and in financial and insurance activities are similar.

Both initial models (*Model 2_B* and *Model 2_C*) have a negative β_0; in this case, when investments in ICT equipments, software and database are at a unitary level (1 million €), the annual employment in this economic activity is -331 people. $\beta_1 = 0.8$, so, for an increase of 1% in the independent variable, the dependent one increases of 0.8%. In the logarithmic model, both coefficients are statistically significant at the 1% level (critical value of 2.58), since $t_{\beta 0} = \frac{0.331}{0.086} = 3.85$ and $t_{\beta 1} = \frac{0.800}{0.012} = 66.67$.

Adding firstly the entity fixed effects and then also the time ones, the model loses this significance: $t_{2\_C} = \frac{0.052}{0.041} = 1.27$ and $t_{3\_C} = \frac{0.013}{0.047} = 0.28$. In *Model 3_C*, an increase of one percentage point in ICT investments corresponds to a growth of 0.052% in annual employment, while in *Model 4_C* the increase is 0.013%.

Even while losing significance, according to R^2, the model improves as the fixed effects are implemented, going from an initial 0.916 to a final 0.998.

- Professional, scientific and technical activities; administrative and support service activities

The final model studied in this thesis is the submodel for professional, scientific and technical activities; administrative and support service activities.

PROFESSIONAL, SCIENTIFIC AND TECHNICAL ACTIVITIES; ADMINISTRATIVE AND SUPPORT SERVICE ACTIVITY

	Log-Log (2_D)	Entity FE (3_D)	Entity+time FE (4_D)
(Intercept)	1.721***		
	(0.076)		
log_ict	0.700***	0.314***	0.084***
	(0.011)	(0.044)	(0.026)
Num.Obs.	380	380	380
R2	0.916	0.995	0.998
R2 Adj.	0.915	0.995	0.998
R2 Within		0.593	0.117
R2 Within Adj.		0.592	0.115

* p < 0.1, ** p < 0.05, *** p < 0.01

Summary 8

Unlike *Model 2_B* and *Model 2_C*, the intercept β_0 is positive, and when the independent variable assumes a unitary value, the logarithm of annual employment is 1.721. In all three models displayed in *Summary 8*, β_1 is positive, so adding fixed effects confirms the positive relationship. These are the consequences of an increase of 1% of investments in ICT equipment, software and database on annual employment: in *Model 2_D* it increases of 0.7%, in *Model 3_D* of 0.314% and in *Model 4_D* of 0.084%. All t-stats are bigger than 2.58 ($\alpha = 1\%$): $t_{2\_D} = 63.63, t_{3\_D} = 7.14$ and $t_{4\_D} = 3.23$.

Model 4_D implies that, checking for omitted variables that vary over years and are stable across countries and, vice versa, omitted variables that vary across countries and are stable over years, the investments in ICT positively affect annual employment.

Once again, the R^2 improves through models, with a final variability explained of 99.8%.

CONCLUSION

This report aims to study the relationship between ICT investments and employment, because ICT has a transformative impact on economies and labour markets. Investments in ICT can lead to the creation of new jobs, especially in technology-driven sectors, while also potentially displacing traditional roles due to automation. By analysing this relationship, policymakers and businesses can anticipate which sectors may experience job growth and which might see reductions, enabling them to develop strategies to mitigate negative impacts. Insights into how ICT investments affect employment in different sectors can help shape policies on education, training, and reskilling. Governments can create programs to upskill workers in fields where job demand is expected to rise due to ICT advancements, reducing unemployment and helping workers transition to new roles. Understanding ICT's effects on employment helps businesses and governments make informed decisions about where to allocate resources to remain competitive in the global market. As ICT adoption often leads to a higher demand for skilled labour, it can exacerbate income inequality and job polarization. Studying the relationship between ICT investments and employment helps identify sectors and groups that may be negatively impacted, allowing for targeted interventions to support equitable economic growth.

The results in *Summary 4* and in the following sub-models display a positive relationship between the investments in the latest technologies and employment, even in sectors in which the literature was more hesitant, but it is important to recognize that this analysis does not consider all possible factors influencing this relationship. While the final model accounts for fixed effects for all three dimensions, other issues remain unresolved in the developed models.

As previously mentioned, there are omitted variables for which data are lacking, for example the level of education and digital skills. Education and workforce skills are key factors that influence how ICT investments translate into employment benefits. Countries with more qualified workers tend to benefit more from the adoption of modern technologies, as a better-prepared human capital can leverage innovations more effectively. The adoption of ICT is often associated with jobs requiring advanced skills, therefore countries with elevated levels of formal education and digital literacy see greater growth in employment in technology-intensive sectors.

Regulation and government policies are also crucial, as they can amplify or limit the effects of ICT investments on employment. A favourable regulatory framework that encourages technology adoption and promotes internet access can facilitate employment growth in technology sectors. These include tax incentives, subsidies for companies investing in modern technologies, and training programs for workers. A flexible labour market that allows companies to adapt quickly to innovative technologies facilitates the creation of jobs related to ICT. However, excessive rigidity in labour laws could slow this process.

It's not just a matter of finding every possible variable; the credibility of the model could also be undermined by reverse causality, where the dependent variable influences the theoretically independent one. For example, it might happen that sectors that have an increase in employees could find in ICT a solution for their management needs. In large companies, human resource management (HRM) systems often better manage payroll, recruitment, training, and employee performance.

Additionally, employing more people in a sector increases competition, driving companies to invest in ICT solutions to gain a competitive edge.

Another limitation of this research is the filters applied; indeed, the selected nations are almost exclusively developed countries, whereas it would be interesting to explore the same topic for the BRIC nations (Brazil, Russia, India, and China), which include two of the countries investing the most in technologies and that have experienced exponential growth in recent times. Alternatively, examining African states that are increasingly investing in this field, such as Libya and Morocco, which are among the most active, or the Seychelles and Mauritius, which have recorded significant increases in digital usage, would be valuable. If nineteen countries are few to make real general considerations, having only four economic activities is also unsatisfactory. It would be necessary to have access to a database providing much more data on every aspect of the economy to deliver a truly comprehensive model. All these shortcomings contribute to undermining the external validity of the discussion.

In conclusion, many authors in the introduction were frightened by a possible job displacement due to the substitution of low skilled workers with automated machines. This study cannot specify if these workers effectively lost their jobs, but what it can underscore is that ICT has created more jobs than it has eliminated; otherwise, there would have been a negative coefficient in the regression models.

Policymakers around the world should continue to invest in ICT in all sectors, keeping in mind that they should also follow a policy of continuous professional development. This will ensure that technologies progress together with the advancement of human skills, preventing a situation in which hiring new workers would be the only way to fully leverage technological innovations.

Gabriele Ferrari

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